NYC 311 SERVICE REQUESTS GRAPH DATABASE PROJECT

Neo4j-Based Analysis

Group Members:

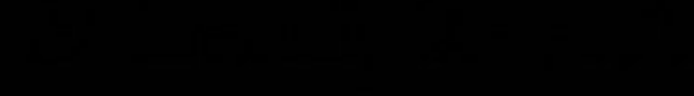
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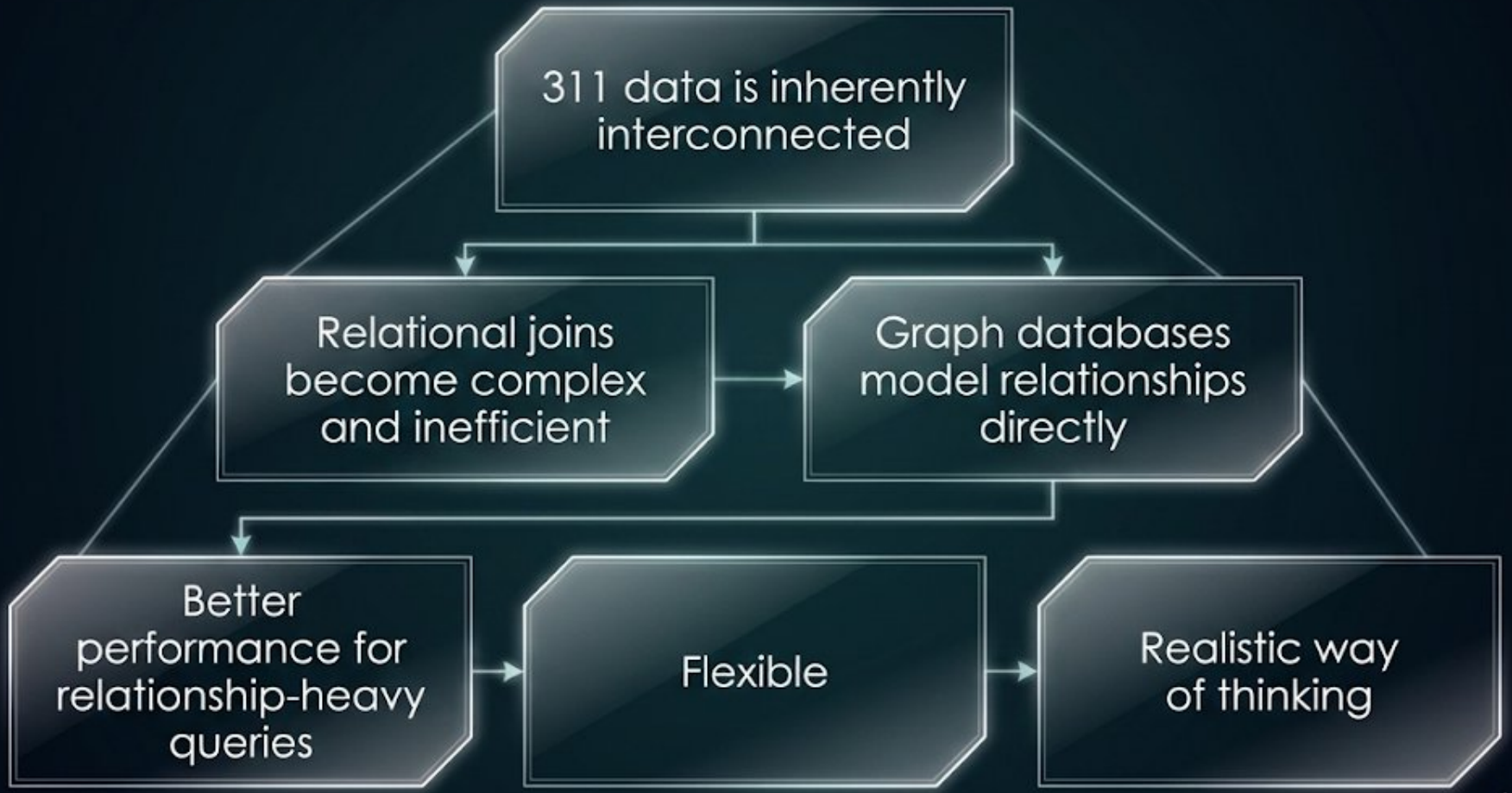
Nobukhosi Sibanda

PROBLEM DEFINITION

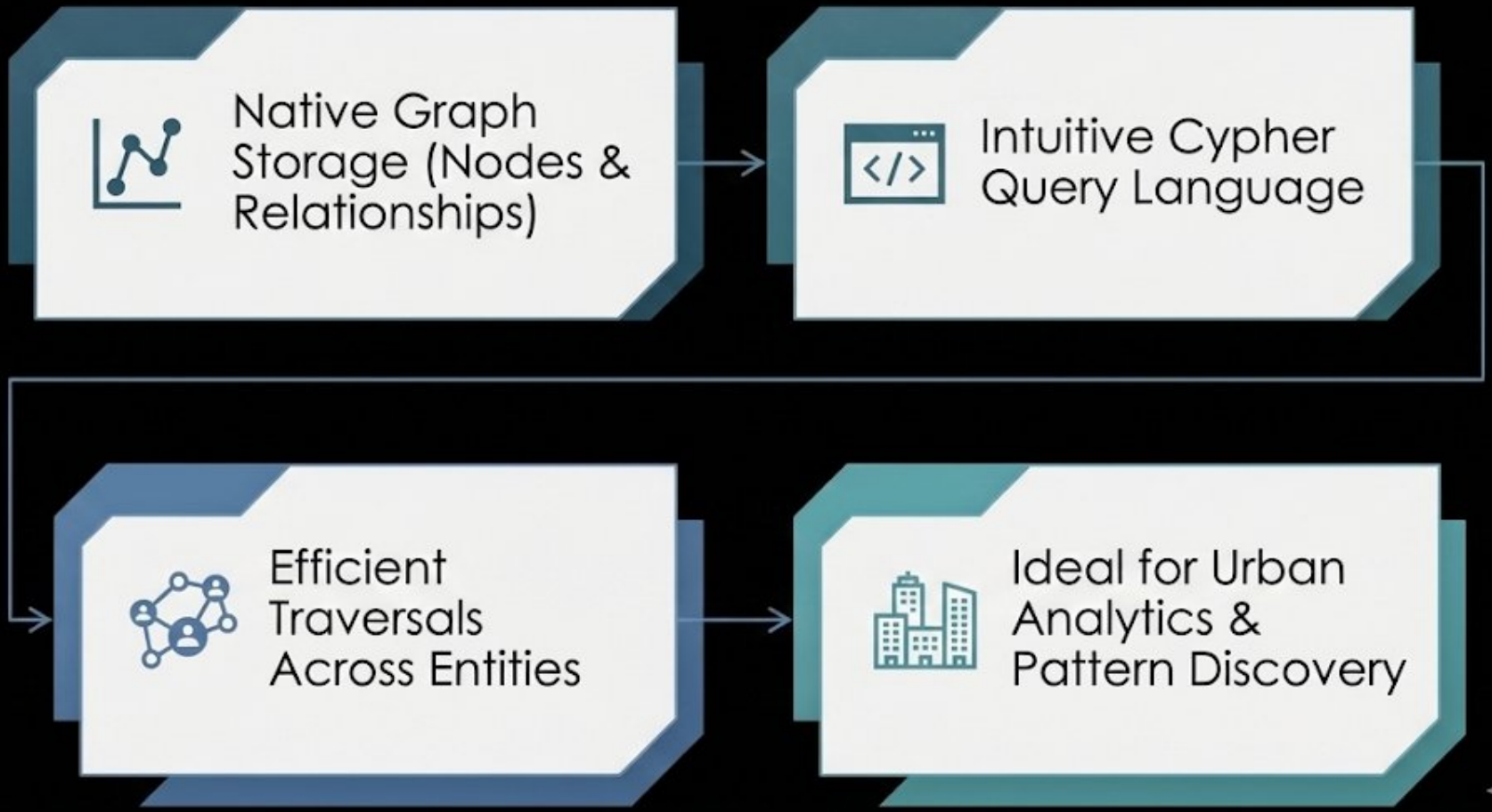
- NYC 311 is the primary non-emergency reporting system
- Handles noise, sanitation, parking, heat, street issues
- Each request connects agency, complaint type, location, and borough
- Goal: Analyze relationships between flat records
- Explore patterns of complaint distribution across boroughs and agencies
- Analyze service performance such as resolution time per agency or request
- Provide a flexible foundation for more advanced analysis in future work



WHY GRAPH DATABASES?



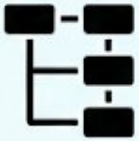
WHY NEO4J?


Native Graph Storage (Nodes & Relationships)
Intuitive Cypher Query Language
Efficient Traversals Across Entities
Ideal for Urban Analytics & Pattern Discovery

DATASET OVERVIEW



NYC 311 SERVICE REQUESTS (2010–PRESENT SUBSET)



305,703 ROWS, 41 COLUMNS



TIME RANGE: NOV 2025 – DEC 2025



MULTIPLE AGENCIES, BOROUGHS, & COMPLAINT TYPES

KEY ENTITIES IDENTIFIED



GRAPH DATA MODEL – NODES

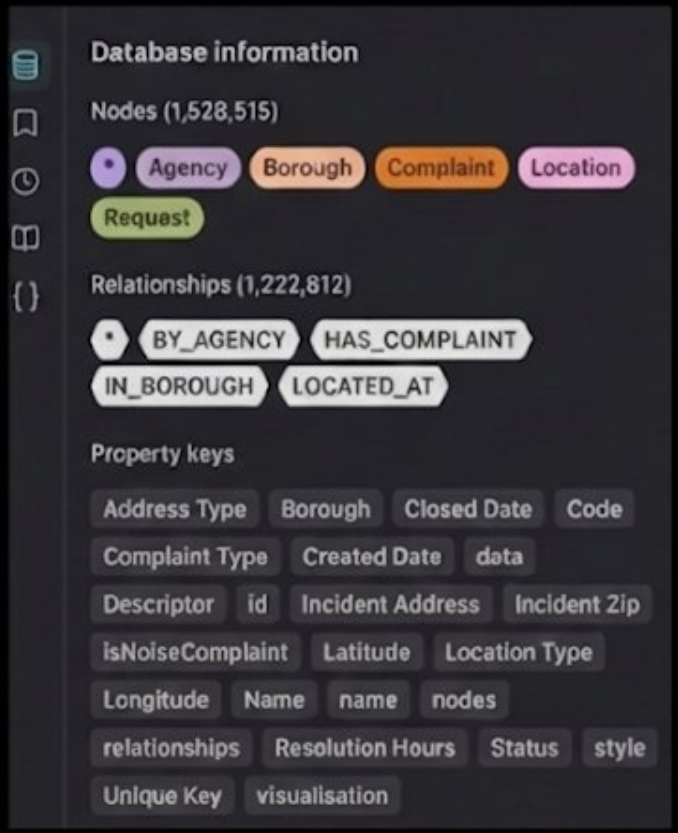


GRAPH DATA MODEL – RELATIONSHIPS



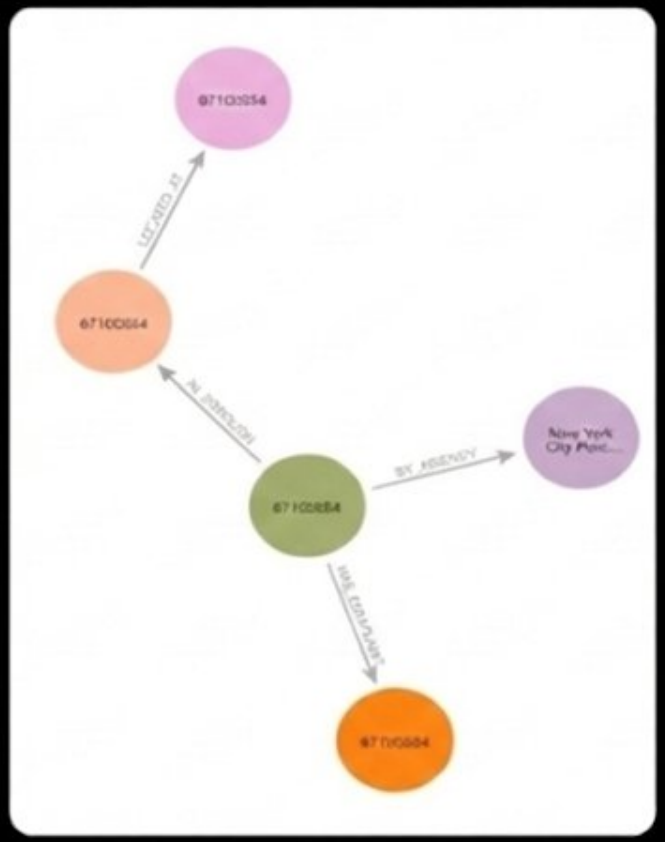
311 Graph Data Model : – Schema & Overview

- **Clean, Simple Model:** Centered on the 311 Service Request.
- **Nodes (Entities)**: Request (Event), Agency (e.g., NYPD), Complaint (Issue Type), Location (Coordinates), Borough (Region).
- **Relationships:** (:Request)-[:BY_AGENCY]->(:Agency), (:Request)-[:HAS_COMPLAINT]->(:Complaint), (:Request)-[:LOCATED_AT]->(:Location), (:Request)-[:IN_BOROUGH]->(:Borough).
- **Key Properties:** Created Date, Status, Incident Address, Descriptor.



Graph Model in Action: Single Request Visualization

- **Visualizing a Request:** This graph shows a single point (ID: 67100554) and its connections.
- **Central Node:** The green node represents the specific Request.
- **Contextual Connections:**
  - Handled BY_AGENCY the New York City Police Department (Purple Node).
  - LOCATED_AT a specific spatial point (Pink Node).
  - IN_BOROUGH [Borough ID] (Light Orange Node).
  - HAS_COMPLAINT type defined by its ID (Dark Orange Node).
- **Benefits:** Allows for highly efficient querying across different dimensions (e.g., find all noise complaints handled by NYPD in Brooklyn).



DATA CLEANSING

- Handled null Closed Date values
- Replaced with 'Not Completed Yet'
- Normalized borough names
- Ensured data consistency

```
1 MATCH (b:Borough)
2 SET b.name = trim(b.name);
```

Get to know the latest in Query|Browser

```
1 MATCH (r:Request)
2 WHERE r.`Closed Date` IN ['Not Completed Yet']
3 RETURN COUNT(r);
4
```

Table RAW

COUNT(r)
43862

Started streaming 1 record after 21 ms and completed after 79 ms.

DATA TRANSFORMATION



Date Object Conversion: Converted Created/Closed Date strings into efficient Date objects.



Resolution Calculation: Derived resolutionHours using advanced APOC algorithms.



Complaint Flagging: Automatically flagged noise-related complaints for analysis.

DATA TRANSFORMATION: TEMPORAL CLEANING & FEATURE ENGINEERING

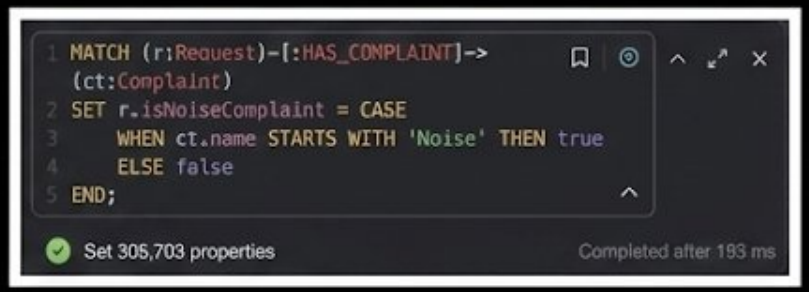
- **Initial State:** 'Created Date' & 'Closed Date' were String format.
- **Action:** Converted to native Date/Datetime objects using apoc.date parse.
- **New Feature:** Derived 'Resolution Hours' for efficiency analysis.
- **Impact:** 785,442 properties updated.

```
MATCH (r:Request)
WHERE r.`Closed Date` <> 'Not Completed Yet'
AND r.`Created Date` IS NOT NULL
  AND r.`Closed Date`  IS NOT NULL
WITH r,
     apoc.date.parse(r.`created Date`,'s','MM/dd/yyyy hh:mm:ss a') AS createdSec,
     apoc.date.parse(r.`Closed Date`,'s','MM/dd/yyyy hh:mm:ss a')  AS closedSec
WHERE createdSec IS NOT NULL AND closedSec IS NOT NULL AND closedSec >= createdSec
SET r.`Created Date`     = datetime({epochSeconds: createdSec}),
    r.`Closed Date`      = datetime({epochSeconds: closedSec}),
    r.`Resolution Hours` = (closedSec - createdSec) / 3600.0;
```

Set 785,442 properties — Completed after 1,378 ms

DATA TRANSFORMATION: DOMAIN CLASSIFICATION (NOISE FLAGGING)

- **Logic:** Graph traversal (Request)-[:HAS_COMPLAINT]-> (Complaint).
- **Action:** Implemented conditional CASE logic to deaminat to identify noise-related complaints.
- **Outcome:** Created 'isNoiseComplaint' (Boolen) flag.
- **Impact:** 305,703 properties classified.



ANALYTICAL QUERIES PERFORMED

- Complaint types citywide
- Noise vs non-noise complaints by borough
- Complaints per borough and agency
- Top agencies by complaint volume

PERFORMANCE ANALYSIS

- Average resolution time per borough
- Average resolution time per complaint type
- Complaints per borough
- NYPD-specific performance analysis

Complaint Types Citywide

Cypher Query

```
MATCH (r:Request)-[:HAS_
COMPLAINT]->
(ct:Complaint)
RETURN ct.`Complaint
Type` AS complaintType,
COUNT(r) AS numRequests
ORDER BY
numRequests DESC
LIMIT 10;
```

Top Complaint Types (Results)

Complaint Type	Number of Requests
Noise - Residential	55220
HEAT/HOT WATER	54275
Illegal Parking	42674
Blocked Driveway	15071
UNSANITARY CONDITION	9321



Noise vs. Non-Noise Complaints by Borough

Cypher Query

```
MATCH (r:Request)-[:IN_BOROUGH]->(b:Borough)
WHERE r.isNoiseComplaint IS NOT NULL
RETURN b.Borough AS borough,
     SUM(CASE WHEN r.isNoiseComplaint THEN 1 ELSE 0 END) AS noiseRequests,
     COUNT(r) AS totalRequests,
     round(100.0 * SUM(CASE WHEN r.isNoiseComplaint THEN 1 ELSE 0 END) / COUNT(r), 2) AS
        pctNoise
ORDER BY pctNoise DESC;
```

Borough Noise Analysis (Results)

Borough	Noise Requests	Total Requests	Percentage of Noise
MANHATTAN	1675	58757	2.85%
BROOKLYN	1335	84479	1.58%
STATEN ISLAND	118	9209	1.28%
QUEENS	761	63701	1.19%
BRONX	481	89357	0.54%
Unspecified	0	200	0.0%

Complaints per Borough and Agency

Cypher Query

```
MATCH (r:Request)-[:IN_BOROUGH]->
  (b:Borough),
  (r)-[:BY_AGENCY]->(a:Agency)
RETURN b.Borough AS borough,
       a.Code AS agencyCode,
       a.Name AS agencyName,
       COUNT(r) AS numRequests
ORDER BY borough, numRequests DESC;
```

Borough & Agency Complaints (Results)		
Borough	**Agency**	**Request**
BRONX	HPD	26004
BRONX	DSNY	19288
BRONX	DEP	4198
BRONX	DOB	3541
BROOKLYN	HPD	26004
BROOKLYN	DSNY	6216
BROOKLYN	DOT	4297
BROOKLYN	DEP	3508
BROOKLYN	DOB	2760
BROOKLYN	DPB	1631
MANHATTAN	HPD	21497
MANHATTAN	NYPD	19288
MANHATTAN	DSNY	3937
MANHATTAN	DEP	3005
MANHATTAN	DOT	2660
MANHATTAN	DHS	2027
QUEENS	NYPD	32900
QUEENS	HPD	12068
QUEENS	DSNY	4840
QUEENS	DOT	4198
QUEENS	DEP	3541
QUEENS	DOB	1720
STATEN ISLAND	NYPD	3104
STATEN ISLAND	DSNY	1479
STATEN ISLAND	DOT	1243
STATEN ISLAND	DEP	1182
STATEN ISLAND	DEP	1066
STATEN ISLAND	DPR	452

Average Resolution Time per Borough

Cypher Query

```
MATCH (r:Request)-[:IN_BOROUGH]->
    (b:Borough)
WHERE r.`Resolution Hours` IS NOT NULL
RETURN b.Borough AS borough,
    COUNT(r) AS numRequests,
    round(AVG(r.`Resolution Hours`),
    2) AS avgResolutionHours
    avgResolutonHours
ORDER BY avgResolutionHours DESC;
```

Borough Resolution Time (Results)

Borough	Number of Requests	Average Resolution Hours
STATEN ISLAND	7858	43.1
Unspecified	108	40.14
MANHATTAN	48173	37.51
BROOKLYN	71671	37.01
BRONX	78322	27.86
QUEENS	55681	24.93

Average Resolution Time for Complaint Type

Cypher Query

```
MATCH (r:Request)-[:HAS_COMPLAINT]->
(ct:Complaint)
WHERE r.`Resolution Hours` IS NOT NULL
RETURN ct.`Complaint Type` AS complaintType,
       COUNT(r) AS numRequests,
       round(AVG(r.`Resolution Hours`), 2)
AS avgResolutionHours
ORDER BY avgResolutionHours DESC
LIMIT 10;
```

Resolution Time by Complaint (Results)

Complaint Type	Number of Requests	Average Resolution Hours
"Taxi Complaint t"	3	560.64
"Taxi Com-pliment"	1	446.13
"For Hire Vehicle Complaint"	129	433.0
"Special Operations "	125	418.94
"Building Drinking Water Tank"	2	313.04
"LinkNYC"	7	258.93
"OUTSIDE BUILDIN NG"	42	249.22

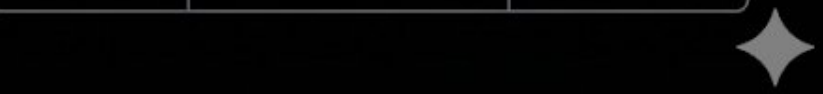
Agencies Handling the Most Complaints Overall

Cypher Query

```
MATCH (r:Request)-[:BY_AGENCY]->
(a:Agency)
RETURN r.Code AS agencyCode,
       a.Name AS agencyName,
       COUNT(r) AS numRequests
ORDER BY numRequests DESC
LIMIT 10;
```

Top Agencies by Complaints (Results)

Agency Code	Agency Name	Number of Requests
NYPD	New York City Police Department	140142
HPD	Department of Housing Preservation and Development	93024
DSNY	Department of Sanitation	19144
DOT	Department of Transportation	14073
DEP	Department of Environmental Protection	12591



Complaints Per Borough

Cypher Query

```
MATCH (r:Request)-[:BY_AGENCY]->
(a:Agency)
RETURN a.Code AS agencyCode,
       a.Name AS agencyName,
       COUNT(r) AS numRequests
ORDER BY numRequests DESC
LIMIT 10;
```

Borough Complaints (Results)

Borough	Number of Requests
“BRONX”	89357
“BROOKLYN”	84479
“QUEENS”	63701
“MANHATTAN”	58757
“STATEN ISLAND”	9209
“Unspecified”	200

Top Complaint Types for NYPD

Cypher Query

```
MATCH (r:Request)-[:BY_AGENCY]->(a:Agency
{Code: 'NYPD'})
MATCH (r)-[:HAS_COMPLAINT]->(c:Complaint)
RETURN c.`Complaint Type` AS complaintType,
       a.Name,
       COUNT(*)           AS numRequests
ORDER BY numRequests DESC
LIMIT 10;
```

NYPD Complaints (Results)

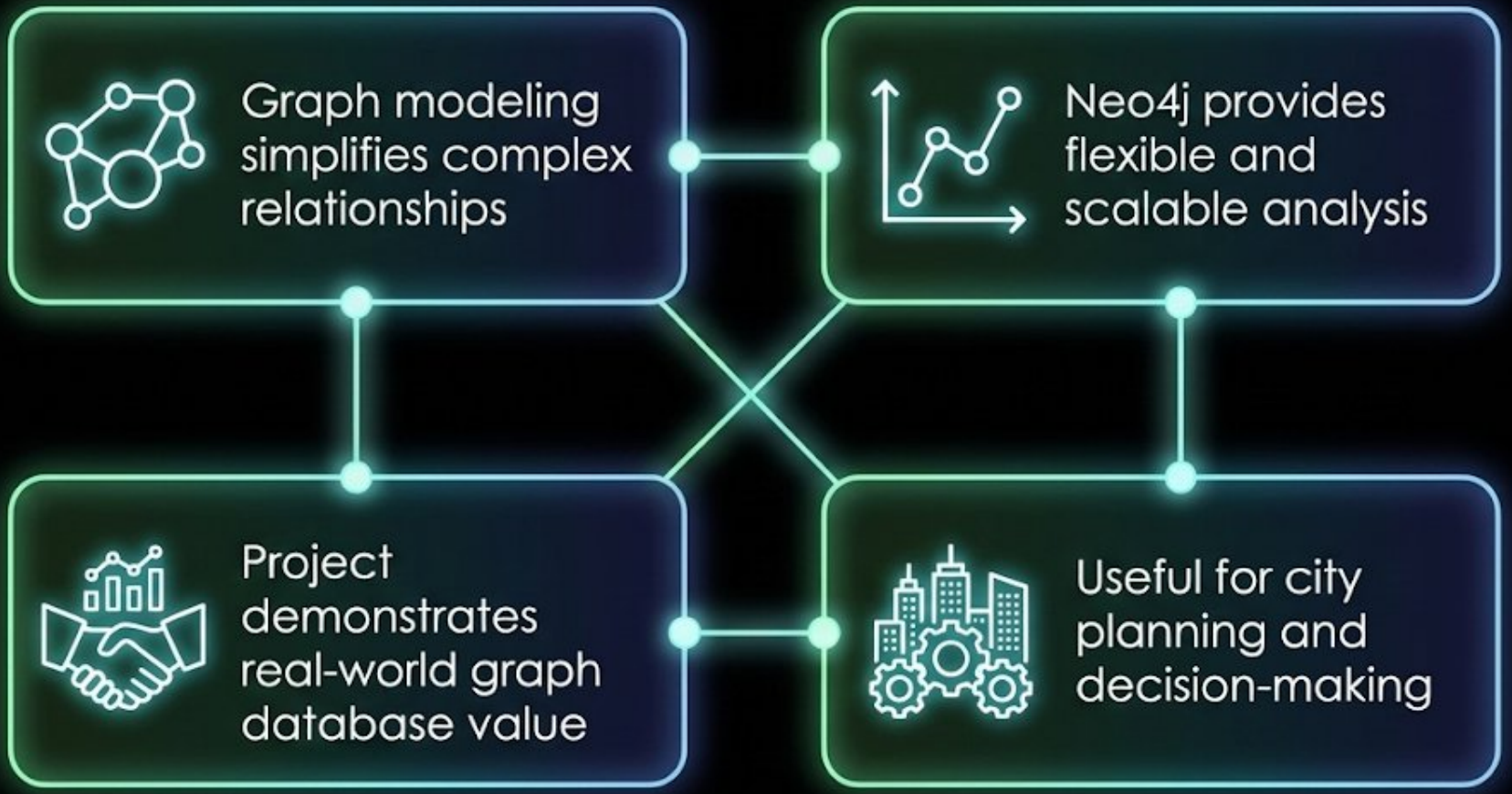
Complaint Type	Number of Requests
Noise - Residential	55220
Illegal Parking	42674
Blocked Driveway	15071
Noise - Street/Sidewalk	6863
Abandoned Vehicle	4708



KEY INSIGHTS ENABLED BY GRAPH MODEL



CONCLUSION


Graph modeling simplifies complex relationships
Neo4j provides flexible and scalable analysis
Project demonstrates real-world graph database value
Useful for city planning and decision-making